

Theorem 3.5. Let U_n be given by (3.11) with $E[h(X_1, \dots, X_m)]^2 < \infty$.

(i) If $\zeta_1 > 0$, then

$$\sqrt{n}[U_n - E(U_n)] \rightarrow_d N(0, m^2 \zeta_1).$$

(ii) If $\zeta_1 = 0$ but $\zeta_2 > 0$, then

$$n[U_n - E(U_n)] \rightarrow_d \frac{m(m-1)}{2} \sum_{j=1}^{\infty} \lambda_j (\chi_{1j}^2 - 1), \quad (3.21)$$

where χ_{1j}^2 's are i.i.d. random variables having the chi-square distribution χ_1^2 and λ_j 's are some constants (which may depend on P) satisfying $\sum_{j=1}^{\infty} \lambda_j^2 = \zeta_2$. ■

We have actually proved Theorem 3.5(i). A proof for Theorem 3.5(ii) is given in Serfling (1980, §5.5.2). One may derive results for the cases where $\zeta_2 = 0$, but the case of either $\zeta_1 > 0$ or $\zeta_2 > 0$ is the most interesting case in applications.

If $\zeta_1 > 0$, it follows from Theorem 3.5(i) and Corollary 3.2(iii) that $\text{amse}_{U_n}(P) = m^2 \zeta_1 / n = \text{Var}(U_n) + O(n^{-2})$. By Proposition 2.4(ii), $\{n[U_n - E(U_n)]^2\}$ is uniformly integrable.

If $\zeta_1 = 0$ but $\zeta_2 > 0$, it follows from Theorem 3.5(ii) that $\text{amse}_{U_n}(P) = EY^2/n^2$, where Y denotes the random variable on the right-hand side of (3.21). The following result provides the value of EY^2 .

Lemma 3.2. Let Y be the random variable on the right-hand side of (3.21). Then $EY^2 = \frac{m^2(m-1)^2}{2} \zeta_2$.

Proof. Define

$$Y_k = \frac{m(m-1)}{2} \sum_{j=1}^k \lambda_j (\chi_{1j}^2 - 1), \quad k = 1, 2, \dots$$

It can be shown (exercise) that $\{Y_k^2\}$ is uniformly integrable. Since $Y_k \rightarrow_d Y$ as $k \rightarrow \infty$, $\lim_{k \rightarrow \infty} EY_k^2 = EY^2$ (Theorem 1.8(viii)). Since χ_{1j}^2 's are independent chi-square random variables with $E\chi_{1j}^2 = 1$ and $\text{Var}(\chi_{1j}^2) = 2$, $EY_k = 0$ for any k and

$$\begin{aligned} EY_k^2 &= \frac{m^2(m-1)^2}{4} \sum_{j=1}^k \lambda_j^2 \text{Var}(\chi_{1j}^2) \\ &= \frac{m^2(m-1)^2}{4} \left(2 \sum_{j=1}^k \lambda_j^2 \right) \\ &\rightarrow \frac{m^2(m-1)^2}{2} \zeta_2. \quad \blacksquare \end{aligned}$$